

VYTAUTAS MAGNUS UNIVERSITY
FACULTY OF INFORMATICS

SOCIAL MEDIA ANALYTICS

Lab work #2

Visualization of Virtual Communities and Explanation of Results

Email-Eu-core Network

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2026

I. Tasks

The objective of this laboratory is to select a real social-network dataset, identify virtual communities within it, visualize their structure, and explain what the detected communities may represent.

The work is divided into three main parts:

- a. Select and Understand the Data
- b. Detect and Visualize Communities
- c. Explain the Results

II. Email-Eu-core Network

1. Dataset and Method

For this laboratory, I selected the Email-Eu-core dataset from the Stanford Large Network Dataset Collection (SNAP). The dataset represents email communication between members of a European research institution. A node represents an individual member of the institution, while a directed edge represents an email relationship between two members, with the direction indicating the sender and recipient.

The original network contains 1,005 nodes and 25,571 directed email relationships. A separate file provides department labels for all 1,005 nodes, which gives additional organizational context for interpreting the detected communities.

I selected this dataset because email communication represents interaction between actors and can therefore be studied as a social network. The department metadata also makes it possible to compare communities detected from network structure with organizational structure.

Dataset source: <https://snap.stanford.edu/data/email-Eu-core.html>

The analysis was carried out in Python using the NetworkX library. For community detection, I used the Louvain method. This method identifies groups with relatively stronger connections within groups and weaker connections between groups, consistent with the course definition of community detection.

The original email network is directed because an email has a sender and a recipient. For Louvain community detection, the network was converted to an undirected graph so that the analysis focused on whether actors were connected rather than on the direction of individual email messages. The original directed network contains 25,571 edges, while the undirected network used for community detection contains 16,706 edges. The reduction occurs because reciprocal relationships between the same pair of actors are represented as a single undirected connection.

2. Main Results and Community Structure

Measure	Result
Nodes	1,005
Original directed edges	25,571
Undirected edges used for detection	16,706
Detected communities	28
Community-level connections	36

The Louvain method identified 28 communities. The communities vary considerably in size. The largest detected community contains 251 nodes, followed by communities containing 145, 129, 129, 111, and 93 nodes. Other larger communities contain between 10 and 64 nodes, while 19 detected communities consist of a single node.

The largest communities are:

Community	Nodes
8	251
28	145
6	129
3	129
2	111
4	93
1	64
16	54
5	10

The visualization of the nine communities containing at least ten nodes provides a clearer view of the main community structure. These communities form a large interconnected structure rather than appearing as completely isolated groups. Several communities have substantial connections with one another, while the single-node communities are peripheral or have limited visible interaction in the simplified view.

3. Important Actors and Connections Between Communities

The community-level analysis identified 36 connections between the 28 detected communities. The strongest observed connection is between Community 8 and Community 28, with 815 connections in the undirected network. Other strong connections include Community 2 and Community 8, with 801 connections, and Community 3 and Community 8, with 725 connections. These results show that the detected communities are interconnected rather than completely separated.

Community pair	Network connections
8 – 28	815
2 – 8	801
3 – 8	725
1 – 8	556
6 – 8	490

Community 8 is particularly well connected to several other communities. The community-level edge values above are network connections in the undirected graph, not counts of the original directed email relationships.

Important actors were examined using degree and betweenness centrality. Degree represents the number of direct connections, while betweenness centrality indicates how often an actor lies on shortest paths between other actors. Node 160 had the highest degree, with 347 direct connections, and the highest betweenness centrality, with a value of 0.087415. It belongs to Community 6 and has connections spanning nine detected communities. Other highly connected actors include Nodes 121, 82, 107, and 86.

Community size and connectivity differences are visible in the analysis, but community density was not calculated separately. Therefore, no direct claim is made about which community is the densest.

4. What the Communities May Represent

The Email-Eu-core network is most closely related to an in-group community. The course describes in-group networks as potentially occurring among professional peers in a work environment. The Email-Eu-core network fits this interpretation because the actors are members of the same research institution and the detected groups are formed from their communication patterns. However, the network does not perfectly represent an in-group structure because several detected communities have substantial connections with one another.

The department labels provide additional evidence for interpreting the communities. Community 4 contains 93 nodes, of which 88 (94.62%) belong to Department 14. Community 3 contains 129 nodes, with 93 (72.09%) belonging to Department 4, and Community 1 contains 64 nodes, with 43 (67.19%) belonging to Department 1. In contrast, Community 8, the largest community with 251 nodes, has only 47 members from its largest department group (18.73%). This indicates that Community 8 is more mixed and does not correspond closely to a single department.

The results suggest that some detected communities may reflect organizational departments or groups of closely interacting colleagues, while others may reflect communication patterns that cross department boundaries. The network provides evidence about interaction patterns, but it does not establish why these communities exist or what was discussed in the emails.

5. Connections Between Communities and Limitations

The cross-community connections indicate that communication occurs between the detected groups. Several actors have connections spanning nine different communities, with Node 160 being particularly notable because of its high degree and betweenness centrality. Such cross-community actors can occupy bridge-like positions in the network and may contribute to interaction between otherwise more separated groups.

The network structure can therefore support an interpretation of potential information flow between communities. However, the network alone cannot establish what information was exchanged, why actors communicated across communities, or whether a connection represents a strong social relationship.

Several limitations should be considered. An email relationship does not necessarily indicate the strength or nature of a social relationship. Louvain communities are produced from network structure and should not automatically be interpreted as formal organizational departments. The full visualization also simplifies a relatively large network, making some individual relationships difficult to observe. Finally, the analysis describes observed network structure but does not establish why particular communities or communication patterns exist.

6. Conclusion

The Email-Eu-core analysis identified 28 communities in a network of 1,005 nodes and 25,571 original directed email relationships. The communities vary substantially in size and are interconnected through 36 community-level connections. Node 160 was identified as a particularly important actor based on its degree, betweenness centrality, and connections across nine communities. Department metadata showed that some communities are strongly associated with particular departments, while others are more mixed. Overall, the analysis demonstrates how community detection, network visualization, centrality measures, and node metadata can be combined to interpret virtual community structure while recognizing the limits of what the network data can establish.

Reference

Stanford Large Network Dataset Collection (SNAP). Email-Eu-core dataset. <https://snap.stanford.edu/data/email-Eu-core.html>

The full-network, main-community, and community-level visualizations are provided in the accompanying Jupyter Notebook. The community-level view uses node size for community size and edge thickness for the number of cross-community network connections.